

Effect of Distance Measures on K-Nearest Neighbour Classifier

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Abstract--Machine learning classifiers vary concerning their learning functions and approach. K-nearest neighbour (K-NN) is a versatile instance-based machine learning algorithm used for classification without building a learning model. Varying distance measures in K-NN for computing distance between instances affect the classification accuracy. Commonly used distance metrics are Euclidean, Manhattan, Minkowski and Mahalanobis. In this work, extensive experiments were carried out on diverse datasets. These datasets vary in terms of the domain, number and types of features. The classification accuracy results obtained by applying different distance measures in the K-NN classifier were compared and analyzed. The results show that the Mahalanobis distance performed better with more than 90% accuracy, than the Euclidean and other measures when applied on most datasets. The other two most significant inferences from this analysis are: first, the Mahalanobis distance considered the correlation between the features and the deviation of instances from the mean, and second, it is scale-invariant.

Keyword--Classification, K-Nearest Neighbour, Euclidean distance, Mahalanobis distance, Manhattan distance, Minkowski distance

I. INTRODUCTION

Instance-based algorithms are supervised machine learning algorithms that classify new instances by calculating distances between these instances and stored classified instances [1]. Classification is a predictive modelling technique where a class label is predicted for new unknown input records based on the learning model built over a set of input records with known class labels from history. Out of many algorithms [2], some are lazy learners who are not actively learning from the training instances. Instead, learning occurs when a relevant query is generated from the new instances of the same type as training points in these algorithms. Some lazy learning algorithms are the Lazy Bayesian rule and K-Nearest Neighbour.

A. K-Nearest Neighbour

The k-Nearest-Neighbours (K-NN) is an effective non-parametric instance-based learning method for a classification task. Learning in this method is done using stored training instances and is activated only when a new data instance is tested for its class category [3]. Then, similar training instances are matched with new instances from the stored training set to predict its class [4]. However, this algorithm involves a lot of complex computations because all the calculations are performed every time the classification task is performed. Furthermore, no learning model is built using training instances to be used later to classify new cases [5].

It is a slow method when applied to data with many instances. The reason behind this slowness is the non-availability of a generalized model in advance. It analyses past data with known class categories whenever a prediction for a new one is needed. There is a bias-variance trade-off in performing K-NN as it has a low bias and high variance. The algorithm of K-NN classification [6] is based on the majority voting of categories of training instances within the decision boundary defined by nearest neighbours [7]. The nearest neighbours of a test instance are identified using distance-based measures.

All nearest training instances in the original K-NN classification are given equal weightage despite their distance from the test instance. Thus, it makes the original uniform K-NN algorithm sensitive to the choice of the value of k. The low value of k makes it more prone to overfitting because of noise in the training instances. This uniform K-NN classifier may provide a false class category to test instances, where most of the training instances are far from the test instance, and only a few are near it. Uniform K-NN classifier is refined by assigning high weights to neighbours nearer to the test instance and low weights to those far from the test instance [8]. It reduces the impact of k on the classification results. The weight associated with k-nearest neighbours is used to do distance-weighted voting for predicting the class category of the test instance. This refined K-NN is a weighted K-NN [9], [10]. The weight (w_i) for the training instance (t_i) is evaluated as the inverse of the square of distance $d(t_{new}, t_i)$ as shown in equation (1)

$$w_i = 1/d(t_{new}, t_i)^2 \quad (1)$$

The main drawback concerning K-NN is selecting the appropriate value of k for precise prediction of the category for new instances. Therefore, the accurate K-NN classification depends on the perfect distance-based measure. Several distance-based metrics can calculate the distance between training and test instance in n-dimensional space.

Distance in machine learning is defined as the closeness and farness of elements from each other in n-dimensional space. Several distance measures [11] are available that satisfy all requirements for an appropriate metric. [12] studied a few distance measures which commonly can be used in the K-NN classification algorithm. [13] Ali & Trundle observed that Euclidean distance was not always the right choice as a measure to calculate the distance in the KNN classification algorithm. A selection of suitable distance metrics helps enhance the classification process's performance.

B. Euclidean distance

Euclidean distance is the most commonly used straight line distance measure in the K-NN classification algorithm [14]. It is a straightforward function to calculate the Euclidean distance for the univariate dataset. It performs well if the features have equal weights and are independent. However, calculating the Euclidean distance between instances for the multivariate dataset is complex due to two main problems. The first problem is that features in a multivariate dataset are represented in various units. Scaling of the features using a z-score is required to overcome the problem. Secondly, the major problem is that there exists a correlation among features. Euclidean distance may deliver misleading results due to these correlated features. Therefore, Euclidean distance is a suitable distance measure to compute the distance between two points only.

C. Manhattan distance

Manhattan distance is generally preferred for high dimensional data. It is more feasible for vectors to be described on a city or chessboard's uniform grid-like blocks rather than a zigzag path. Generally, Manhattan distance is preferred instead of Euclidean distance for two vectors in an integer feature space [15].

D. Mahalanobis distance

The Mahalanobis distance (MD) is the distance measure to find the distance between a data instance and a distribution in a multidimensional space. It was introduced by P. C. Mahalanobis, an Indian scientist, in 1936. It is a multidimensional generalization for measuring the standard deviations of the test instance t_{new} from the mean of training instances distribution. Features are represented by axes drawn at right angles to each other for Euclidean distance calculation. Therefore, the straight distance between any two data points in Euclidean space can be measured using the straight distance formula. However, the axes are not necessarily placed at the right angle if data points show some correlation among each other. Therefore, using the distance formula, it isn't easy to measure the straight distance between data instances. The problem regarding the measurement of distances between correlated data instances in multiple dimensional spaces is resolved by Mahalanobis distance [16]. For uncorrelated data instances, the Euclidean distance is the same as the Mahalanobis distance. Therefore, Mahalanobis Distance [17] is preferred in cases where correlation exists among features of data instances. This is explained using the mathematical formula for Mahalanobis distance (equation 2).

$$d_{mb}(t_{new}, \bar{t})^2 = (t_{new} - \bar{t})^T C^{-1} (t_{new} - \bar{t}) \quad (2)$$

where, t_{new} is a test instance with feature vector $(f_{i1}, f_{i2}, f_{i3} \dots f_{in})$.

$\bar{t}$: arithmetic mean of training instances having mean feature vector $(\bar{f}_1, \bar{f}_2, \bar{f}_3 \dots \bar{f}_n)$.

C^{-1} : inverse covariance matrix of independent feature variables.

$(t_{new} - \bar{t})$ is the distance of the test instance vector from the mean vector. Dividing this factor by the covariance

matrix C is a multivariate equivalent of the standard z-score given as equation (3) for sample or population.

$$z = (x_i - \bar{X})/\sigma \quad (3)$$

The covariance is high when the Mahalanobis distance is computed for the strongly correlated data instances. Dividing by a high covariance in equation (2) will reduce the distance. Alternately, if the data instances are not correlated, covariance is small, and the distance computed does not decrease considerably. Another drawback of Euclidean distance is that the scaling of features data is required to provide consistent results. The Euclidean distance varies for the same feature data if represented in different units. The Mahalanobis distance re-scaled the feature values to have unit variance, and it corresponds to standard Euclidean distance in the transformed multivariate space. Therefore, the Mahalanobis distance is unitless and scale-invariant [18].

Mahalanobis distance does three major transformations to determine the distance, as shown in Fig. 1 effectively: first, it converts the correlated feature values into the uncorrelated form. Second, feature values are scaled to formulate variance equal to 1. And finally, it calculates the Euclidean distance [19].

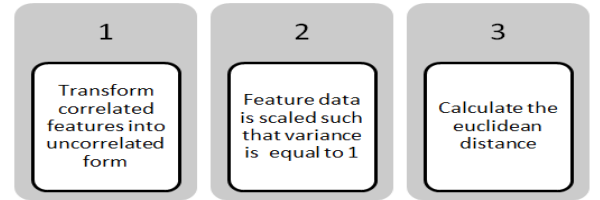


Fig. 1. Mahalanobis computation process

E. Minkowski distance

The Minkowski distance was defined by the mathematician from Germany, Hermann Minkowski. It is a generalized form of the Euclidean and the Manhattan distance measure. It measures the distance between two data instances in a normed vector space. Moreover, Minkowski distance is sensitive to the scale of feature values compared to scale-invariant Mahalanobis distance.

II. RELATED WORK

K-NN classifier algorithm computes the distance between a test instance and similar training instances [20]. The distance between the test and the training instances is arranged in sorted order. The first k training instances are taken for the class category's majority voting. Thus, the class category found in a majority in the k-nearest neighbour set is allocated to the test instance.

There are two significant points of consideration for improving the performance of the K-NN classifier for the test instance: First, the correct choice of the distance formula to be used to calculate distance between the two instances and second, choosing the optimum value of k. Their review paper [21] analyzed the K-NN classifier's performance (quantified using accuracy, precision, and recall) using several distance measures. The experiments were conducted on different types of real-world datasets. The results show that the performance of the K-NN classifier varies remarkably on the distance measure used for classification [22]. [23] state that

the neighbourhood size k strongly affects the K-NN classification performance for mainly small sample sizes. They proposed a generalized mean distance-based K-NN classifier using multi-generalized mean distances to minimize the effect of k on the classifier performance. [24] examine the feasibility of using only one fixed k value for Fuzzy-KNN on each testing sample.

Several standards and modified distance measures [25] are available for measuring the distance between the instances. The most commonly used are Euclidean, Manhattan, Minkowski and Mahalanobis. Euclidean distance is the most frequently used, which calculates the straight distance between the two instances. [26] applied many distance measures on synthetic datasets with different distributions to record the classifier performance. They inferred that the choice of distance measure for a typical dataset has a massive impact on classifier performance. Euclidean distance does not consider any data characteristic while computing the distance between instances [27]. The computation of distance must comprise the effect of aspects of data such as the scale of data, the relationship between features and how far the data from the mean is.

In this research article, 20 real-world datasets with varying characteristics were taken to study the effect of various distance measures on the performance of K-NN. The data is explored by studying its mean, median, standard deviation, correlation etc. The experiments were conducted by applying the K-NN classifier algorithm with four distance measures, namely, Euclidean, Manhattan, Minkowski and Mahalanobis, and varying the values of k . 5-fold cross-validation was applied. To calculate the accuracy, five different training and test sets were taken to undergo the classifier algorithm separately. The accuracy for a particular dataset is the average of their five rearranged training test sets accuracies.

III. DATA SKELETON

For this analysis, 20 datasets of different domains with nominal, ordinal and categorical data were downloaded from the UCI [28] and Kaggle [29] repositories. The primary characteristics of all the datasets are shown in Table I. The multivariate datasets were explored regarding the number of features, characteristics, and types. Box plots showing statistics associated with features of datasets are used to visualize the explorations. The box plot depicts the x-axis as a range of values and the y-axis as features. Every feature is analyzed with the help of a box plot showing its median, percentile, minimum and maximum value and distribution of values. These high and low dimensional datasets were grouped into categorical, numerical, and mixed categories and a few boxplots are shown in Fig 2. The analysis of these boxplots conveys that the datasets are statistically very different. A lot of outliers are visible in the boxplot for the wholesale customer, yeast and diabetes dataset. The seed dataset's features are well distributed within the range of the data domain. Many outliers in the number of features are noticed in the boxplot of the ionosphere and glass dataset as well. The boxplot of the thyroid allbp dataset illustrates the range of continuous and nominal features domains with most of the outliers found in continuous features. Similar observations are there in other datasets also.

TABLE I: CHARACTERISTICS OF THE DATASETS

Dataset	No of record	Type	Target Var #	Attributes #	
				Numeric #	Categorical #
Breast Cancer	684	Categorical	2	0	10
Dermatology	366	Categorical	6	1	34
Primary Tumor	336	Categorical	22	0	16
Car	1728	Categorical	4	6	1
Zoo	102	Categorical	7	16	1
Heart	271	Mixed	2	5	9
HCV	1385	Mixed	4	20	9
Thyroid	2800	Mixed	5	6	21
Immunology	90	Mixed	2	5	3
Wholesale customer	440	Mixed	2	6	2
Diabetes	769	Numerical	2	8	1
Transfusion	749	Numerical	2	4	1
Yeast	1484	Numerical	10	8	1
Seed	210	Numerical	3	7	1
Sonar	208	Numerical	2	60	1
ionosphere	351	Numerical	2	34	1
Glass	214	Numerical	7	9	1
Haberman	306	Numerical	2	3	1
Leaf	340	Numerical	36	15	1
Wine quality	1600	Numerical	6	11	1

IV. DATA EXPLORATION

The first step of data exploration empowers data analysts to detect abnormalities visually and identify relationships between various features. Then, data visualization and statistical techniques are used to describe dataset characteristics in the form of its size, accuracy, range, and standard deviation.

Data description of the datasets were analyzed; Standard deviation in data description for the feature shows how data

is distributed around the mean. The standard deviation of data in all the features of the breast cancer dataset is small, which signifies that information is less dispersed over the mean, as shown in Fig. 3. Minor variation from mean also infer that outlier is not affecting the mean value, and chances of outliers are less in the data value of features. The outliers are easily visible in all features except the region and channel of wholesale customer data by observing the range of values. For example, the minimum value of feature grocery in this dataset is three. The maximum is 92780; this high range also contributes to a falsely high mean and standard deviations value. Also, it is not describing the behaviour of actual data correctly.

The heatmaps, a data visualization tool, demonstrate the relationship between pair of features with colour variants to exhibit a correlation coefficient. The heatmap showing relationships and data description including mean, percentile, the maximum and minimum value was generated using tools in Python. After calculating the correlation coefficient matrix for all feature pairs of datasets, a heatmap was created to exhibit the magnitude of the coefficient by variation in colours. The heatmap shows a mirror image around the diagonal, indicating that feature pairs are not affected by their order. The diagonal elements offer a value of 1, which is the variance of the particular feature. Since the heatmap of some of the datasets has many features, Fig. 4 shows some of the heatmaps of datasets with a smaller number of features. The value and colour are shown at the intersection of each feature cell at the x and y coordinate, showing its coefficient of correlation in the heatmap. The feature intersection cell having a value equal to zero indicates the pair of features are not correlated. The threshold for the coefficient of correlation is taken as 0.1. Therefore, cells at the intersection of features have values greater than 0.1 and less than -0.1, taken as a positive and negative correlation. The rest pair of features are less correlated or not correlated. For example, in Fig. 4 diabetes dataset heatmap, feature pairs (age, pregnancies) have a correlation coefficient of 0.5, indicating a positively correlated. In contrast, feature pairs (age, BMI) having a correlation coefficient of 0.04 implies no correlation among each other. Also, the glass dataset heatmap shows negatively correlated feature pairs one of them with a correlation coefficient of -0.04 is Barium (Ba) and potassium (K). For example, all features are correlated in the breast cancer dataset, whereas features are not related in the car and Haberman dataset.

V. EXPERIMENTAL FRAMEWORK

After exploring the data of three different category datasets, numerical, categorical, and mixed features, it was inferred that datasets vary. The datasets were pre-processed by converting categorical attributes into numeric using the integer coding method. Also, instances with missing values in some datasets were removed. Training and test sets were created using the holdout method to build a learning model and test it. The K-NN measuring classifier with different distance measures was applied to the training set to fit the model. Distance between data points in the K-NN classifier was calculated in both ways: uniform and weighted. A kernel function for the weighted K-NN classifier is taken as an inverse distance function whose value decreases as the

distance increases. An experiment was repeated for different values of k ranging from 1 to 70 to find the set of k values that produce maximum accuracy. Five seed values were taken to select the training and test data randomly. All models generated with varying k and several distance measures were tested using test data classification results when applied to the trained model. The predicted class values of test data were compared with original class categories to find the classification accuracy with the learning model. Average accuracy was calculated from the accuracy calculated for five training and test data sets. Fig. 5 shows the experimental flow of operation.

VI. RESULTS AND DISCUSSION

The K-NN classifier with distance measures Euclidean, Minkowski, Manhattan and Mahalanobis was applied on training data with k ranging from 1 to 70. Accuracy was calculated from the predicted value of test data for all the datasets and the maximum accuracy for particular k values was taken for study. The average accuracy for all the datasets with varying seed values for uniform and weighted distance kernel functions was tabulated. The accuracies calculated for the categorical datasets are shown in Table II. The accuracy calculated using Mahalanobis distance in K-NN classifier is maximum in uniform and weighted K-NN classifier for three categorical datasets breast cancer, car and zoo, out of five categorical datasets. The breast cancer dataset has all feature pairs correlated and all are slightly dispersed from their mean, showing high accuracy compared to other distance measures used. Car dataset with features having low standard deviation and no correlation among features also performed well with accuracy of 97.57% and 97.80% for uniform and weighted kernel functions respectively using Mahalanobis distance measure in the K-NN classifier. The thyroid dataset consisting of all features with low standard deviation also shows great accuracy values for all distance measures. Overall accuracy is very low in HCV, with a high standard deviation for a few features. The performance measure accuracy computed for datasets having mixed type features is shown in Table III. Out of all mixed-type feature datasets, heart, HCV, thyroid and wholesale customers have maximum accuracy when Mahalanobis distance is used to build the K-NN model. A significant difference in accuracy is observed in the heart dataset, having 58% correlated feature pairs and a low standard deviation for features. The performance of the K-NN classifier using Mahalanobis distance is also efficient for numerical datasets.

Table IV shows five out of 10 numeric datasets, giving the highest accuracy in uniform and weighted K-NN using Mahalanobis. Furthermore, the K-NN classifier with Mahalanobis distance measure performed well on 12 high and low dimensional datasets with varying types of features and behaviour. Mahalanobis distance considers the correlation among the features to calculate the distance of test data points from the distribution of training data points. Mahalanobis distance is unitless and scale-invariant. Therefore, it can give an outstanding performance on datasets with varying features.

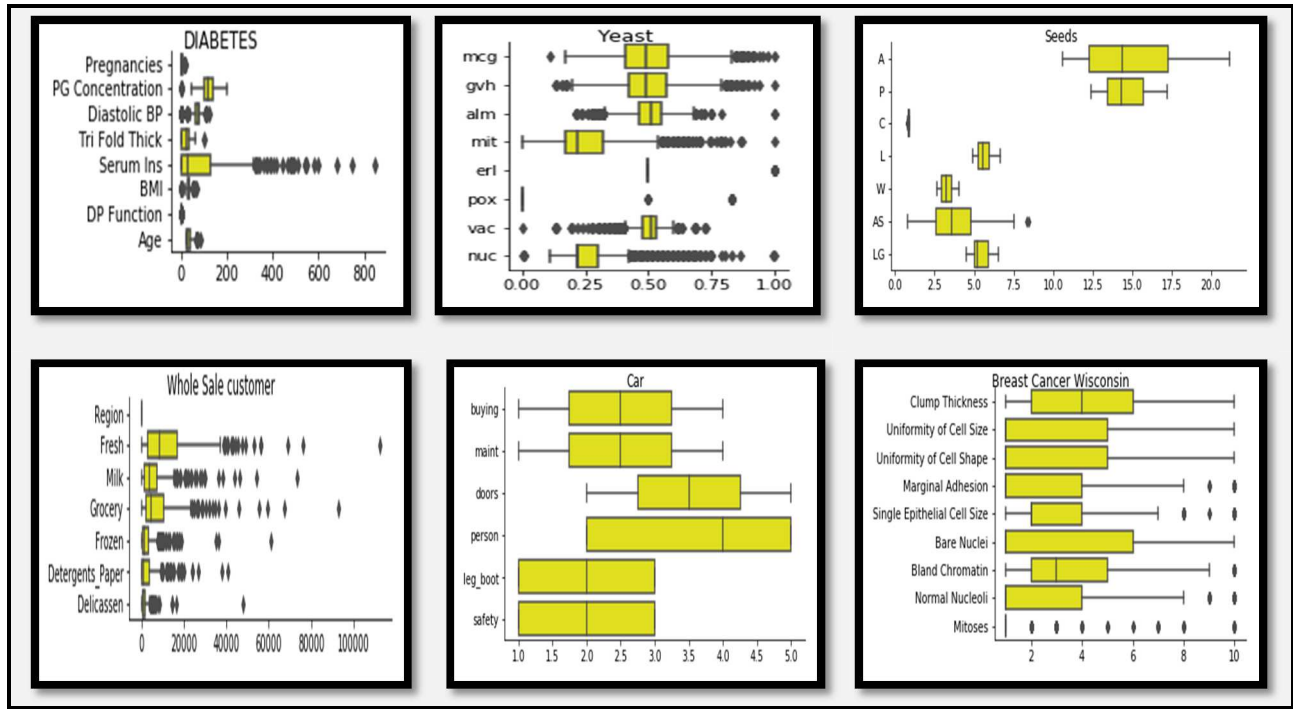


Fig. 2. Statistical Description Boxplots



Fig. 3. Standard Deviation of features in a different dataset

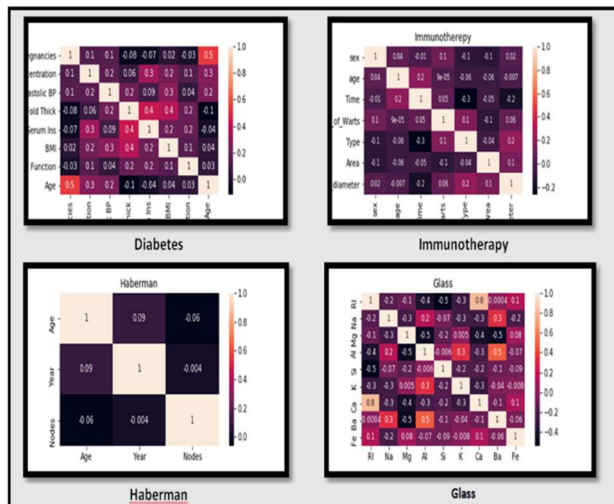


Fig. 4. Heatmaps showing the correlation between pairs of features

For example, K-NN classifier with Mahalanobis distance applied to leaf dataset with features having low standard deviation and 76% correlation. The result is 74.41% and 76.47% accuracy for uniform and weighted kernel functions, respectively. A similar trend appeared in the

Wine quality dataset. On the other hand, the Haberman dataset with no correlation feature pairs and a high standard deviation in most of its features, Mahalanobis gives 77% accuracy, which is less than Minkowski and Euclidean.

TABLE II: PERFORMANCE MEASURES FOR CATEGORICAL DATASETS

Dataset	Distance Measures	Accuracy(%)	
		Uniform	Distance
Breast Cancer	Euclidean	68.61	66.72
	Minkowski	68.61	66.72
	Mahalanobis	95.77	95.91
	Manhattan	68.91	67.73
Dermatology	Euclidean	90.00	90.83
	Minkowski	85.83	86.11
	Mahalanobis	89.44	88.33
	Manhattan	96.11	96.94
Primary tumour	Euclidean	46.47	42.94
	Minkowski	46.47	42.94
	Mahalanobis	46.47	40.88
	Manhattan	46.47	41.76
Car	Euclidean	96.24	96.64
	Minkowski	96.24	96.70
	Mahalanobis	97.57	97.80
	Manhattan	96.12	96.82
Zoo	Euclidean	94.29	96.19
	Minkowski	93.33	95.24
	Mahalanobis	96.19	96.19
	Manhattan	94.29	96.19

TABLE III: PERFORMANCE MEASURES FOR MIXED DATASETS

Dataset	Distance Measures	Accuracy(%)	
		Uniform	Distance
Heart	Euclidean	68.89	68.15
	Minkowski	69.26	69.63
	Mahalanobis	85.93	85.93
	Manhattan	74.08	72.96
HCV	Euclidean	30.47	28.80
	Minkowski	30.90	29.17
	Mahalanobis	29.75	30.54
	Manhattan	29.16	28.67
Thyroid	Euclidean	62.68	62.96
	Minkowski	62.50	62.61
	Mahalanobis	71.46	71.86
	Manhattan	64.04	64.32
Immuno-therapy	Euclidean	80.00	78.89
	Minkowski	83.33	78.89
	Mahalanobis	85.56	84.44
	Manhattan	83.33	82.22
Wholesale Customer	Euclidean	92.50	92.50
	Minkowski	91.81	91.81
	Mahalanobis	88.40	88.87
	Manhattan	92.05	92.27

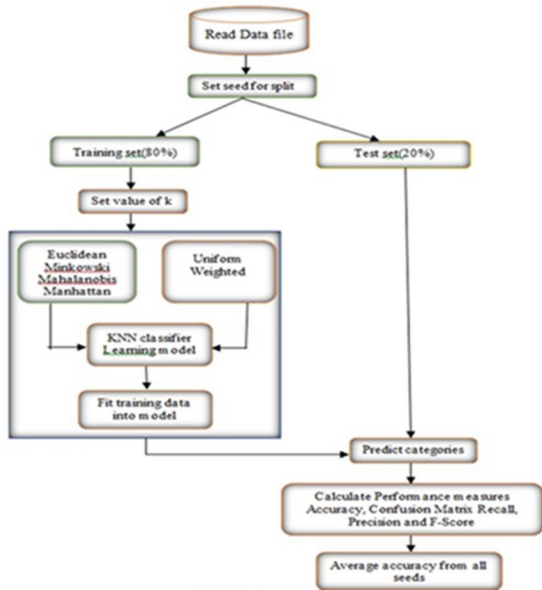


Fig 5: Experimental Flow of operations

VII. CONCLUSION AND FUTURE WORK

The study was conducted to compare the impact of different distance measures on the accuracy of the K-NN classification algorithm. The twenty multivariate datasets of varying nature were used to find the accuracy of this analysis. The accuracy results from this analysis shows that the performance of the K-NN classifier using the Mahalanobis distance measure is better than the other distance measures used. The multivariate datasets with categorical, numerical and mixed type features were

tested with five different subsets of training and test data to avoid overfitting.

TABLE IV: PERFORMANCE MEASURE FOR NUMERICAL DATASETS

Dataset	Distance Measures	Accuracy(%)	
		Uniform	Distance
Diabetes	Euclidean	77.01	76.36
	Minkowski	75.32	75.19
	Mahalanobis	77.01	77.40
	Manhattan	77.53	77.40
Transfusion	Euclidean	77.07	77.33
	Minkowski	77.33	77.33
	Mahalanobis	79.47	78.27
	Manhattan	77.20	76.93
Yeast	Euclidean	62.70	63.97
	Minkowski	62.29	63.91
	Mahalanobis	62.22	63.63
	Manhattan	63.10	64.85
Seed	Euclidean	91.42	91.42
	Minkowski	91.90	90.48
	Mahalanobis	97.62	97.62
	Manhattan	93.81	92.86
Sonar	Euclidean	86.67	86.20
	Minkowski	84.29	84.29
	Mahalanobis	76.67	78.10
	Manhattan	87.62	87.62
Ionosphere	Euclidean	90.57	87.71
	Minkowski	87.71	86.86
	Mahalanobis	84.86	81.43
	Manhattan	91.43	90.00
Glass	Euclidean	70.70	70.70
	Minkowski	72.10	72.56
	Mahalanobis	64.65	69.30
	Manhattan	71.16	72.56
Haberman	Euclidean	78.36	78.69
	Minkowski	79.02	78.36
	Mahalanobis	77.05	76.07
	Manhattan	77.70	78.03
Leaf	Euclidean	57.06	57.06
	Minkowski	55.00	55.88
	Mahalanobis	74.41	76.47
	Manhattan	62.06	63.53
Wine Quality	Euclidean	57.63	66.57
	Minkowski	57.13	66.00
	Mahalanobis	64.94	72.38
	Manhattan	58.69	67.81

It is established that even if the features are correlated in multivariate datasets, the K-NN classifier with Mahalanobis distance measure performed better in terms of accuracy of classification. This correlation among features is transformed into a covariance matrix that is used to calculate Mahalanobis distance between test and training instances distribution in multidimensional space. In contrast, the Euclidean distance calculates the straight line distance ignoring the correlation among features and sometimes giving misleading results. It was observed that the K-NN classifier using Mahalanobis distance performed better even if the dataset's feature domain values were not normalized. This analysis can be extended to determine the performance of these distances measures in the K-NN classifier for text data.

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